

Rail Grinding Machine Water Subsystem

Model-Based Systems Engineering Case Study using Arcadia and Capella

Executive Summary

This project demonstrates the application of the Arcadia Model-Based Systems Engineering (MBSE) methodology using Capella for the development of a Rail Grinding Machine Water Subsystem.

The subsystem is responsible for providing continuous water availability for grinding operations, crew support activities, dust suppression, fire and spark hazard control, and emergency response. The study covers the complete architecture lifecycle from Operational Analysis through System Analysis, Logical Architecture, Physical Architecture, and end-to-end traceability.

The objective of this project was to establish a structured architecture that ensures safe, reliable, and continuous water availability throughout rail grinding operations while maintaining traceability from stakeholder requirements to physical implementation.

1. Problem Statement

Rail grinding operations generate heat, dust, sparks, and potential fire hazards during grinding activities. Continuous water availability is essential to maintain grinding quality, control grinding wheel temperature, suppress dust, mitigate fire and spark hazards, and support onboard crew activities.

The water subsystem must ensure reliable water storage, distribution, monitoring, refilling, and emergency handling while supporting safe operation throughout the grinding mission.

2. Project Objectives

The primary objectives of the subsystem are:

- Provide continuous pressurized water for rail grinding operations.
 - Maintain safe grinding wheel temperatures.
 - Suppress dust generated during grinding activities.
 - Control fire and spark hazards.
 - Supply water for onboard crew requirements.
 - Detect low water conditions.
 - Support emergency shutdown and recovery.
 - Maintain traceability from requirements to physical implementation.
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3. Operational Analysis (OA)

Operational Analysis was performed to understand the operational environment, stakeholders, operational capabilities, and operational activities that justify the existence of the water subsystem.

3.1 Operational Entities

The primary operational entities identified were:

- Crew
- External Water Supply
- Environment

3.2 Operational Capabilities

The following operational capabilities were identified:

- Ensure Continuous Rail Grinding Operation
- Maintain Safe Grinding Wheel Temperature
- Ensure Fire and Spark Safety
- Maintain Acceptable Dust Levels
- Sustain Crew Operational Readiness

3.3 Operational Activity Breakdown

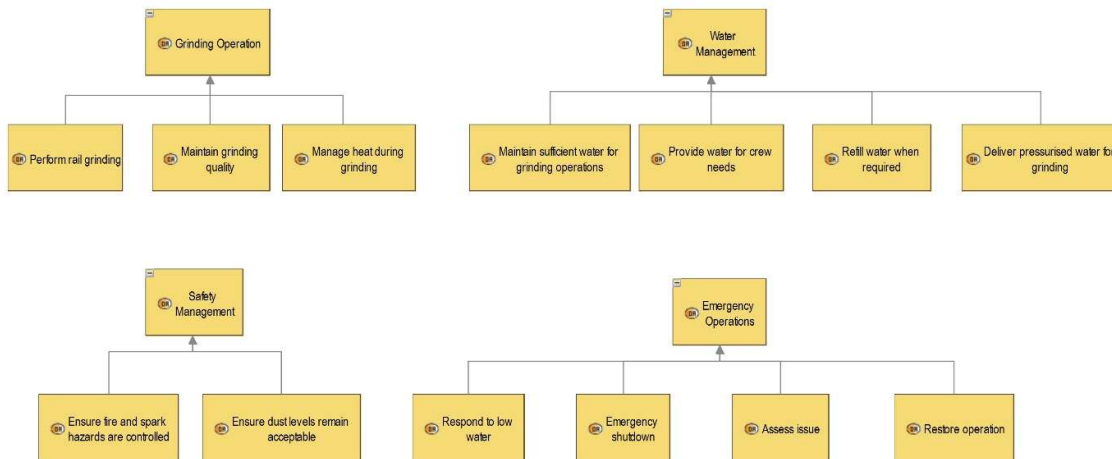


Fig 1: [OABD] Root Operational Activity

The Operational Activity Breakdown Diagram identifies the major operational activities necessary to achieve the operational capabilities.

3.4 Operational Activity Interaction

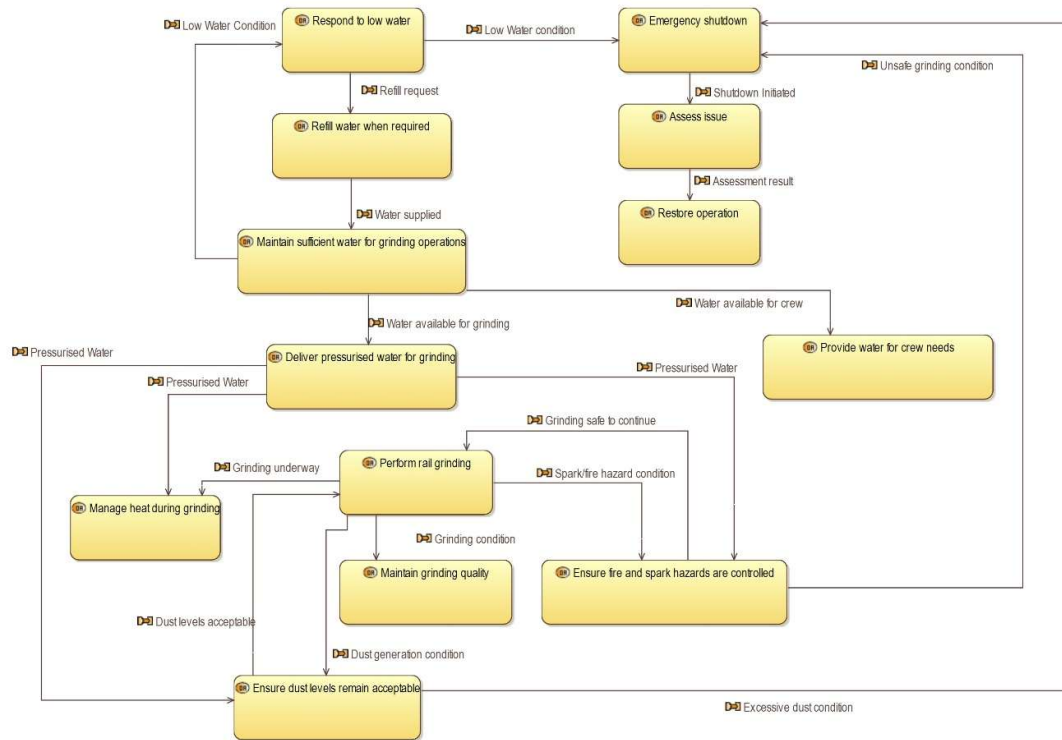


Fig 2: [OAIB] Water System Operations

The Operational Activity Interaction Diagram defines the interactions and exchanges between operational activities.

3.5 Operational Scenario

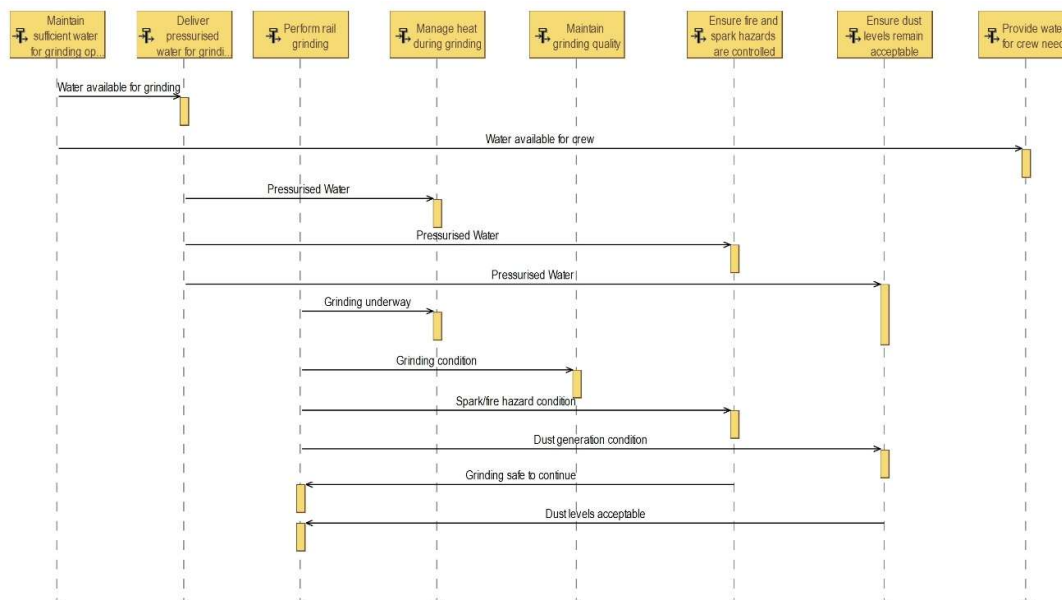


Fig 3: [OAS] Ensure continuous rail grinding operation

The Operational Scenario validates the operational flow of events during grinding activities.

4. System Analysis (SA)

System Analysis transforms operational needs into system responsibilities and functions.

4.1 System Context

The System Context Diagram identifies the external actors interacting with the Rail Grinding Machine Water Subsystem.

4.2 System Functions

The primary system functions identified include:

- Supply Grinding Water
- Monitor Water Level
- Perform Water Refill
- Detect Low Water Condition
- Supply Water to Crew Facilities
- Regulate Grinding Cooling
- Ensure Dust Levels Remain Acceptable
- Ensure Fire and Spark Hazards are Controlled
- Emergency Shutdown
- Assess Issue
- Restore Operation

4.3 System Functional Breakdown

The System Functional Breakdown Diagram decomposes system responsibilities into functional elements.

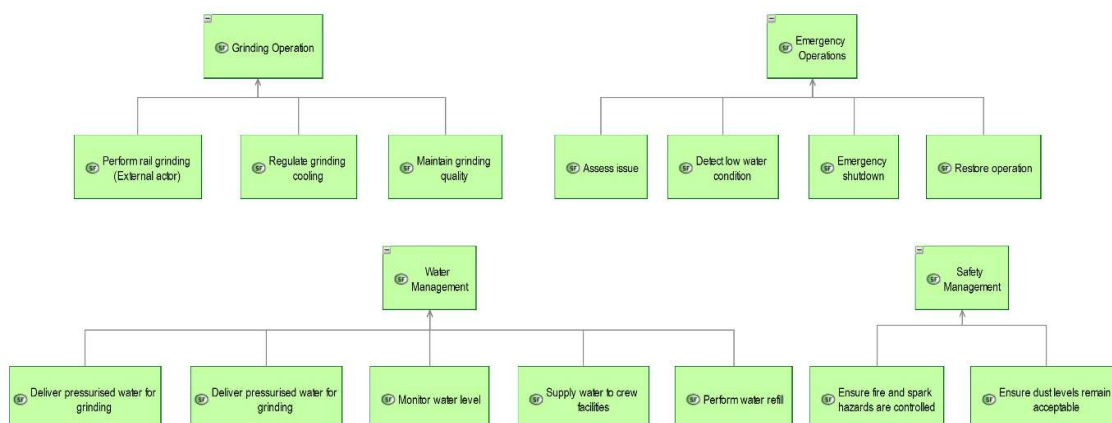


Fig 4: [SFB] Root System Function

4.4 Functional Exchanges

The System Data Flow Diagram defines the functional interactions and exchanges between system functions.

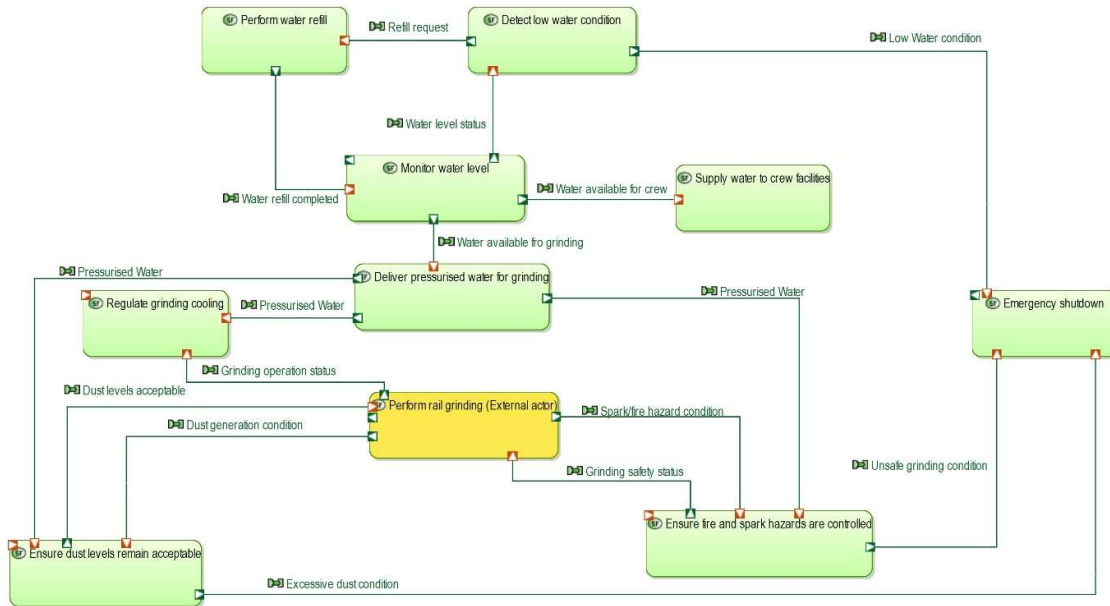


Fig 5: [SDFB] Root System Function

4.5 Modes and States

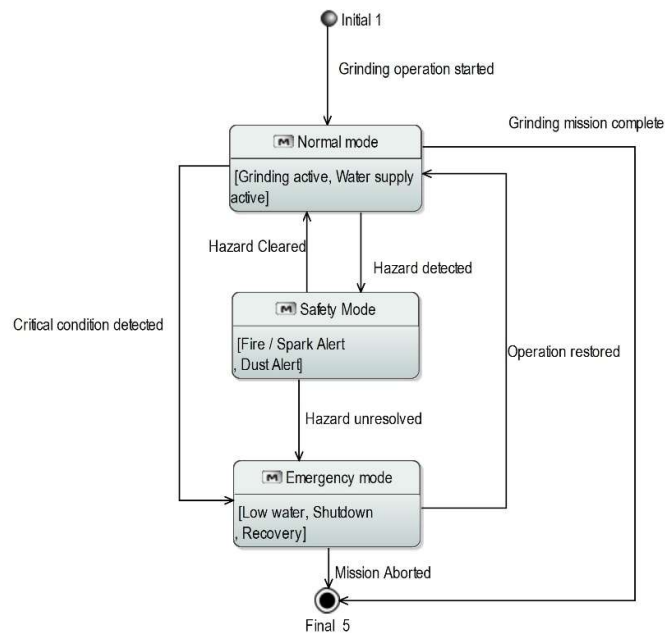


Fig 6: [MSM] Operational Analysis

The Mode State Machine defines the operational modes and transitions of the subsystem.

The identified modes are:

- Normal Mode
- Safety Mode
- Emergency Mode

5. Logical Architecture (LA)

Logical Architecture groups system functions into logical responsibilities independent of implementation.

5.1 Logical Components

The logical architecture consists of the following logical components:

- Water Management Unit
- Cooling Unit
- Hazard Suppression Unit
- Crew Water Supply Unit
- Monitoring and Control Unit
- Emergency Control Unit

5.2 Logical Component Breakdown

The Logical Component Breakdown Diagram defines the logical structure of the subsystem.

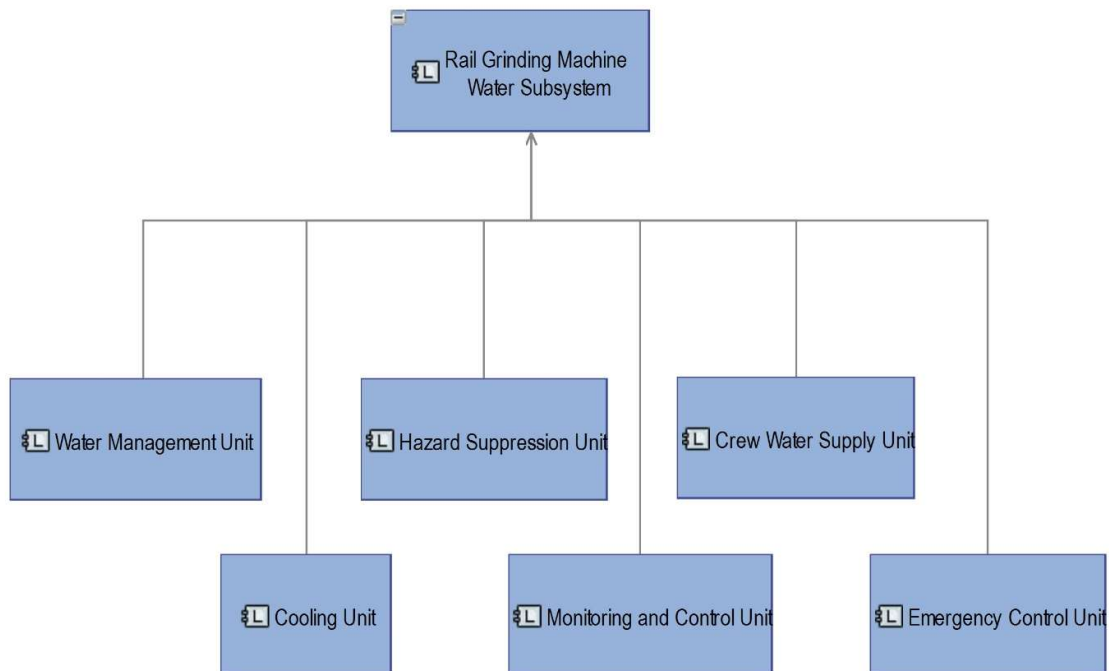


Fig 7: [LCBD] Structure

5.3 Logical Architecture Allocation

The Logical Architecture Blank Diagram allocates system functions to logical components.

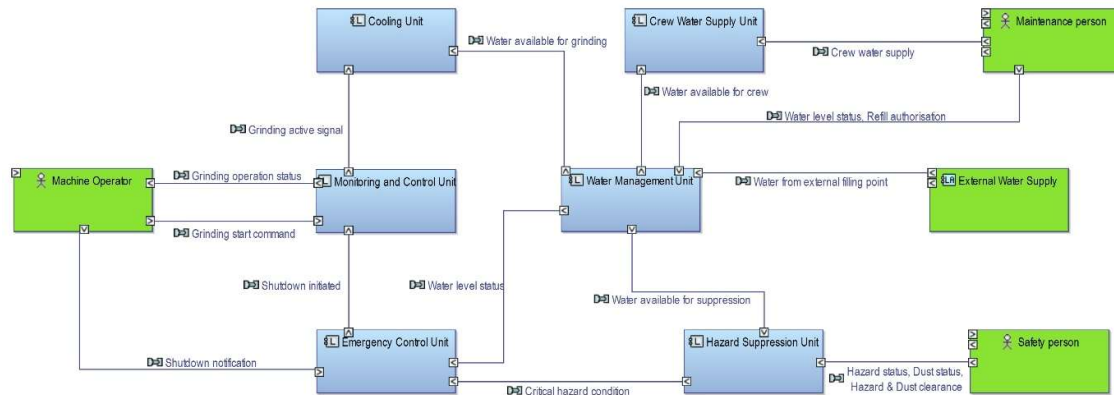


Fig 8: [LAB] Structure

Key Architectural Decisions

Separation of Cooling and Hazard Suppression

Cooling and hazard suppression functions were separated because they serve different operational objectives and utilize different delivery mechanisms.

Dedicated Emergency Control Unit

Emergency functions were separated from normal monitoring and control functions to isolate safety-related behaviour from routine operation.

6. Physical Architecture (PA)

Physical Architecture realizes the logical architecture through actual physical assemblies and equipment.

6.1 Physical Components

The subsystem consists of the following major physical assemblies:

Grind Car Assembly

- Control and Sensing Assembly
- Spray Bar Assembly
- Tie Spray Assembly
- Hose Reel Assembly

Camp Car Assembly

- Main Water Tank Assembly
- Water Cannon Assembly
- Hose Reel Assembly

- Crew Distribution Assembly
- Overhead Water Storage Assembly
- External Filling Point

6.2 Physical Component Breakdown

The Physical Component Breakdown Diagram defines the physical decomposition of the subsystem.

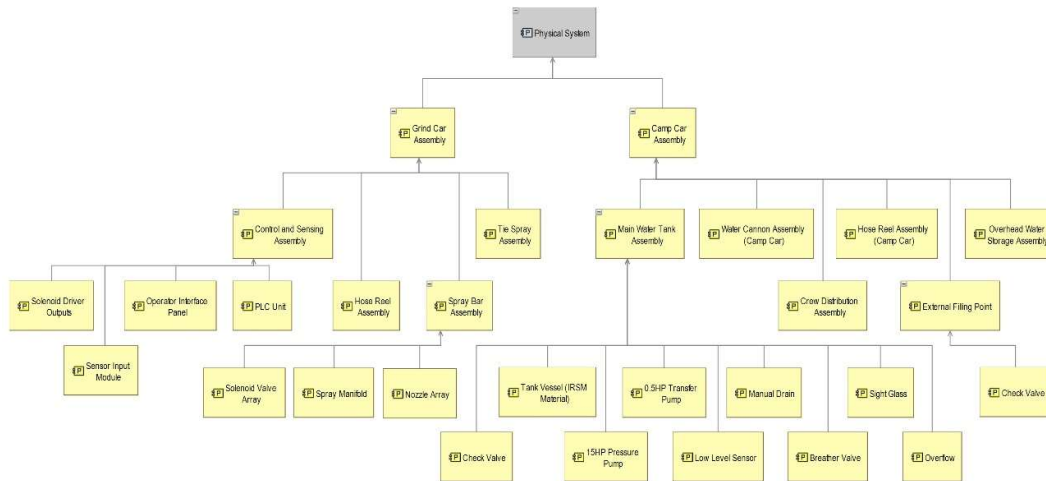


Fig 9: [PCBD] Structure

6.3 Physical Interfaces

The Physical Architecture Blank Diagram defines the interfaces and exchanges between physical components.

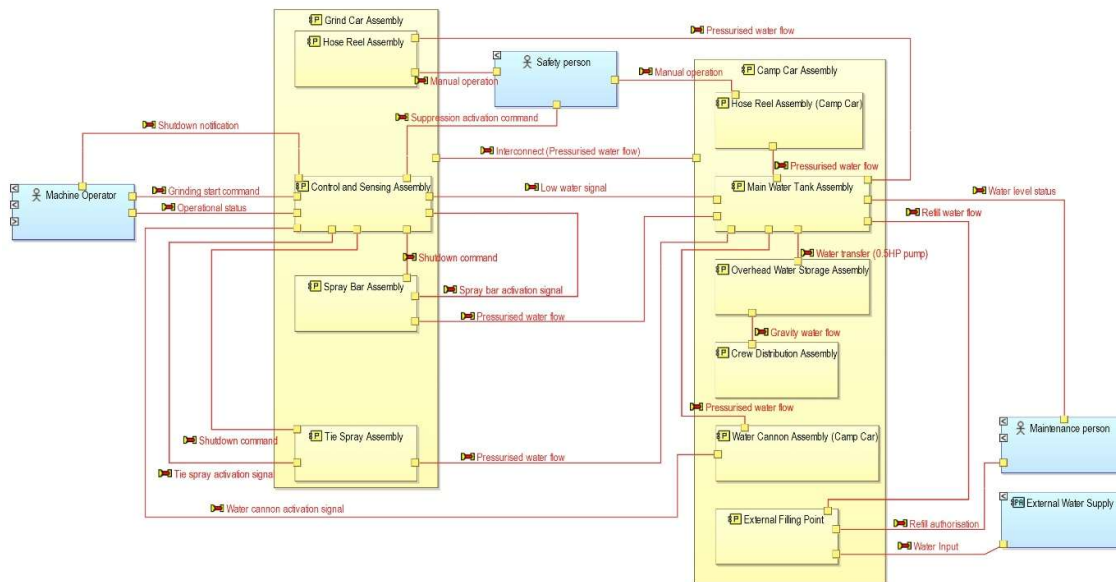


Fig 10: [PAB] Structure

7. Requirements Traceability

The project maintains complete traceability between requirements and implementation.

Traceability chain:

- Requirement
- Operational Activity
- System Function
- Logical Component
- Physical Component
- Verification Method
- Verification Status

SOURCE	OA ACTIVITY	SA FUNCTION	LA COMPONENT	PA COMPONENT	STATUS	VERIFICATION METHOD	VERIFICATION STATUS
OA	Deliver pressurised water for grinding	Deliver pressurised water for grinding	Water Management Unit	Main Water Tank Assembly (15HP Pressure Pump)	FULLY TRACED	TEST	PLANNED
OA	Ensure fire and spark hazards are controlled Ensure dust levels remain acceptable	Ensure fire and spark hazards are controlled Ensure dust levels remain acceptable	Hazard Suppression Unit	Hose Reel Assembly/Water Cannon Assembly (Camp Car)/Hose Reel Assembly (Camp Car)	FULLY TRACED	DEMONSTRATION	PLANNED
OA	Manage heat during grinding	Regulate grinding cooling	Cooling Unit	Spray Bar Assembly/Tie Spray Assembly	FULLY TRACED	TEST	PLANNED
OA	Provide water for crew needs	Supply water to crew facilities	Crew Water Supply Unit	Crew Distribution Assembly	FULLY TRACED	DEMONSTRATION	PLANNED
OA	Respond to low water	Detect low water condition	Monitoring and Control Unit	Control and Sensing Assembly	FULLY TRACED	TEST	PLANNED
OA	Maintain sufficient water for grinding operations / Refill water when required	Monitor water level/Perform water refill	Water Management Unit	Main Water Tank Assembly/External Filling Point	FULLY TRACED	INSPECTION/ANALYSIS	PLANNED
OA/ENG	Provide water for crew needs / Refill water when required	Supply water to crew facilities/Perform water refill	Water Management Unit	Overhead Water Storage Assembly/External Filling Point	FULLY TRACED	INSPECTION/ANALYSIS	PLANNED
OA/ENG	Emergency shutdown	Emergency shutdown	Emergency Control Unit	Control and Sensing Assembly	FULLY TRACED	TEST	PLANNED

Fig 11: Requirement Traceability Matrix

This traceability ensures that every requirement is realized, allocated, and verifiable.

8. Verification Strategy

Verification methods used in the project include:

- Inspection
- Analysis
- Functional Testing
- Demonstration

Verification status is maintained within the Requirements Traceability Matrix.

9. Lessons Learned

The project provided several key MBSE insights:

- Operational activities must remain independent of system implementation.
- System functions represent responsibilities of the system rather than operational needs.
- Logical architecture should be based on responsibility allocation rather than physical equipment.
- Physical architecture provides realization of logical responsibilities.
- Traceability is critical for maintaining consistency across architecture levels.

- Arcadia provides a structured framework for architecture development from concept to implementation.
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10. Conclusion

This project successfully demonstrates the application of the Arcadia MBSE methodology to the Rail Grinding Machine Water Subsystem.

The architecture was developed through Operational Analysis, System Analysis, Logical Architecture, and Physical Architecture while maintaining complete traceability from stakeholder requirements to physical implementation.

The resulting architecture provides a structured, traceable, and scalable foundation for the development and verification of onboard water management systems used in rail grinding operations.